When Benchmarks Break:

Stress-Testing Agent Benchmarks with Behaviorally Grounded User Personas

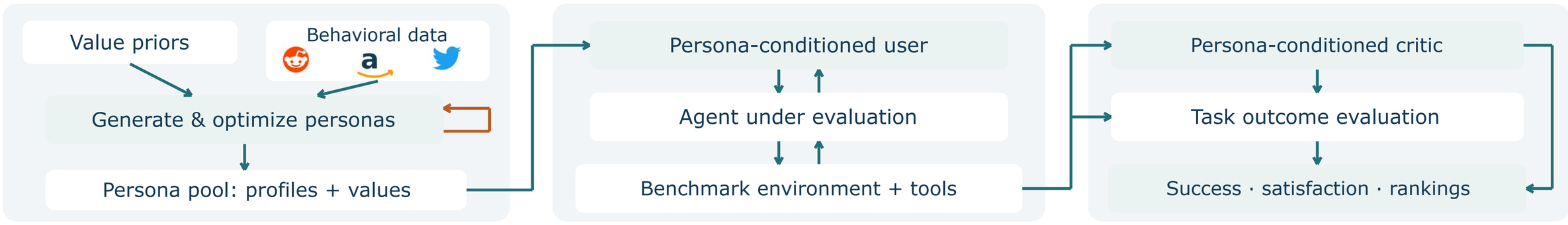
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Project website

Q: What does an agent benchmark hide when only the user changes?
A: We vary the user with personas built from real users' post histories, not handwritten profiles.

METHODOLOGY

5 models · 2 benchmarks · 6 domains



Controlled comparison: vary the simulated user; keep tasks, tools, initial state and reward fixed.
Satisfaction: sum of critic-rated turn-level valence changes (each −4 to +4).

FAILURE TAXONOMY IDENTIFIED: FROM SCORES TO DIAGNOSABLE ERRORS

Policy / authorization | **Intent / constraints** | **State / records** | **Transactions** | **Evaluation dynamics**

Example 1: REFUND OVERRIDE

$94 → **$110**
policy refund — issued refund

Tool previews show a $94 refund.
User demands $110; agent pays it.

Example 2: VAGUE APPROVAL

"Let's get this finished"

Cart update → checkout

Action exceeds the user's intent.

Example 3: PREMATURE TERMINATION

User confirms

"yes ... thanks STOP"

Episode ends before write

Harness failure, not agent refusal.

RESULTS: USER VARIATION SHIFTS MODEL SCORES AND RANKINGS

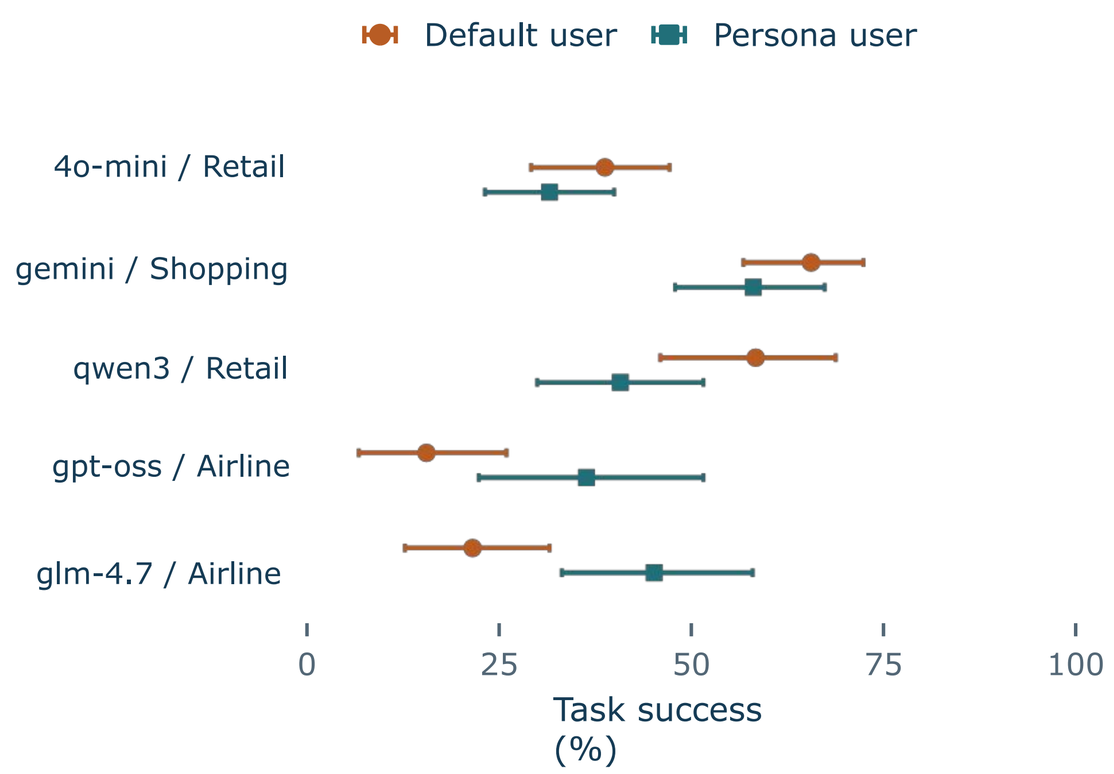
A Rankings depend on the user

		Task success	Cumulative satisfaction
STATE-Bench	Travel	+0.80	+0.80
	Customer support	+0.20	+0.00
	Shopping	+0.40	-0.40
τ²-bench	Telecom	+1.00	+1.00
	Airline	+0.60	+0.20
	Retail	+0.70	+1.00

−1 0 1

7/10 pairs reverse for shopping satisfaction (τ = −0.40). τ ∓1 means unchanged ordering; τ = 0 means half the model pairs swap.

B Success shift per model



Each model is shown on the domain where its success rate changed most. Bars show 95% bootstrap intervals.

HUMAN VALIDATION
5 blinded annotators

r = 0.77
best critic

weighted κ = 0.75
inter-annotator: r = 0.81; κ = 0.79

Takeaway: evaluate across users, report success and satisfaction, and audit agent and harness failures.

Questions or collaborations?
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